

Haedi DeAngelis Thelen

University of Pennsylvania
Blockley Hall 423 Guardian Drive, Philadelphia, PA 19104
(505) 206-1246
Haedi.Thelen@PennMedicine.UPenn.edu

Education	Doctor of Philosophy - Epidemiology University of Pennsylvania, PA Matriculated	Fall 2022
	Doctor of Pharmacy University of New Mexico, Albuquerque, NM	2022
	Master of Arts, Bioethics Trinity Graduate School, Deerfield, IL	2015
	Bachelor of Arts, Mathematics Trinity College, Deerfield, IL	2015
Research Experience	PhD Student Research Post Doctoral Pharmacoepidemiology Fellow Department of Biostatistics and Epidemiology University of Pennsylvania, Philadelphia, PA	2022 - Current
	Research Interests: treatment effect heterogeneity, causal inference, drug associated kidney injury Advisor: Sean Hennessy, PharmD, PhD Co-Advisor: Todd Miano, PharmD, PhD - Evaluate trends in NSAID use and treatment effect heterogeneity of NSAID associated acute kidney injury in hospitalized patients using electronic healthcare database records - Develop risk modeling approaches to obtain individualized estimates of ibuprofen associated acute kidney injury in hospitalized patients using observational data	
	Graduate Student Intern Sandia National Laboratories, Albuquerque, NM - Worked with computer science, mathematics, and public health experts to model COVID-19 vaccine distribution and allocation scenarios to aid decision makers - Conceptualized and won a COVID-19 internally funded research project to examine gamma irradiation sterilization of N95 respirators; worked with radiation, aerosol, and material science experts to design experiments, and published in peer reviewed journal - Trained machine learning models to identify cattle with respiratory syndromes in drone footage to support Global Security internally funded research - Collaborated with Los Alamos National Lab to analyze Veterans Affairs databases and develop predictive machine learning models for mental health applications	2019-2022

Laboratory Technician Mercier Infectious Disease Lab

2018-2022

UNM College of Pharmacy, Albuquerque, NM

- Researched and published on the synergy between vancomycin and piperacillin/tazobactam against methicillin-resistant *Staphylococcus aureus*
- Designed and executed experiments investigating synergy between cefixime and amoxicillin/clavulanate against ESBL-producing *Escherichia coli*
- Extracted and analyzed clinical data from patient health records to assess vertical HIV transmission risk among pregnant women with HIV
- Lab responsibilities included experimental design, proposal submission, data analysis, abstract and manuscript drafting, and poster presentations

Research Rotation

2021-2022

TriCore Reference Lab, Albuquerque, NM

- Developed machine learning model to predict chronic kidney disease progression from a seven-year, longitudinal dataset collected from Southwest United States
- Utilized Pandas and NumPy Python packages for pre-processing data and Microsoft PowerBI's AutoML technology for machine learning ensemble model development
- Continued post rotation efforts focused on improving model and publishing results in a peer-reviewed journal

Undergraduate Research Fellow

Summer 2014-Fall 2015

Santa Fe Institute, Santa Fe, NM

- Worked with Dr. Benjamin Althouse using computer modeling techniques to design, code, and implement a dynamic transmission model of *Bordetella pertussis* with numerical solving approaches of ordinary differential equations
- Evaluated cost-effectiveness ethical implications of the different vaccine strategies

Computer Skills

Python for data analytics and machine learning applications, R, Mathematica, LaTeX, and high-performance computing for large datasets and machine learning

Lab Skills

Time kill assays, broth microdilution assays, E-tests, DNA extraction and isolation, polymerase chain reactions, and gel electrophoresis

Publications

1. Dhaubhadel, S, Ganguly, K, Ribeiro, RM, **Haedi Thelen**, McMahon, BH, et al., High dimensional predictions of suicide risk in 4.2 million US veterans using ensemble transfer learning. *Scientific Reports*. 2024 January. DOI: 10.1038/s41598-024-51762-9.
2. Hernandez, BN, Dilworth, TJ, Kesner, J, Ryan, K, **Haedi Thelen**, Mercier, RC. MIC discrepancies between parenteral and oral anti-staphylococcal beta-lactams among MSSA. *Chemotherapy*. 2023 March. DOI: 0.1159/000526630.
3. Klise K, Beyeler W, Erin Acquesta, **Haedi Thelen**, Makvandi M, and Finley P. Prioritizing vaccination based on analysis of community networks. *Applied Network Science*. 2022 December. DOI 10.1007/s41109-022-00522-7.
4. **Haedi DeAngelis**, Dilworth TJ, Mercier RC. Examining the combination of cefixime and amoxicillin/clavulanate against extended-spectrum beta-lactamase (ESBL)-producing *Escherichia coli* isolates. *Chemotherapy*. 2022 May. DOI: 10.1159/000524707.
5. **Haedi DeAngelis**, Grillet A, Nemer M, Wasiolek M, Hanson D, Omana M, Sanchez A, Vehar D, Thelen P. Gamma radiation sterilization of N95 respirators leads to decreased respirator performance. *PLOS One*. 2021 April. DOI: 10.1371/journal.pone.0248859.

6. Dilworth T, Sanchez DA, Anderson BA, **Haedi DeAngelis**, Mercier RC. 2019. Exploring the role of piperacillin and tazobactam in combination with vancomycin against methicillin-resistant *Staphylococcus aureus*. *Chemotherapy*. 64.5-6: 233-237.
7. **Haedi DeAngelis**, Scarpino SV, and Althouse BM. 2016. Modeling the Effects of Priming With the Whole-Cell *Bordetella Pertussis* Vaccine—Reply. *JAMA Pediatrics*. 170(12):1229-1229.
8. **Haedi DeAngelis**, Scarpino SV, Fitzpatrick MC, Galvani AP, and Althouse BM. 2016. Epidemiological and economic effects of priming with the whole-cell *Bordetella pertussis* vaccine. *JAMA Pediatrics*. 170(5):459-65.

Professional Presentations

1. **Haedi Thelen**, Keele LJ, Miano, TA. *Accepted for March 2024*. Risk of Acute Kidney Injury in Hospitalized Patients Treated with Short-Courses of Ibuprofen: a Retrospective Cohort Study. Oral Presentation at the International Society of Pharmacoepidemiology Mid-Year Meeting, Orlando, FL.
2. Ali R, **Haedi Thelen**, Dodd M, Borunda T, Mithiwi A, Koenig M, Argyropylos C. 2021. Evaluating Progression of Chronic Kidney Disease via Lab Data Using Artificial Intelligence. Poster presentation at the American Society of Health Systems Pharmacist Annual Conference, online presentation.
3. **Haedi Thelen**. 2021. Observational Studies of COVID-19 Vaccines. Presented at 2021 COVAX Scholars UNM Health Sciences Center, Albuquerque, NM.
4. **Haedi DeAngelis**, Mercier RC. 2019. Examining the Combination of Cefixime and Amoxicillin/Clavulanate Against Extended-Spectrum Beta-Lactamase (ESBL)-Producing *Escherichia coli* isolates. Poster presentation at American College of Clinical Pharmacy Annual Meeting, New York, NY.
5. **Haedi DeAngelis**, Scarpino SV, Fitzpatrick MC, Galvani AP, and Althouse BM. 2016. The Economic and Ethical Impact of Utilizing a Priming Dose of Whole-Cell *Bordetella pertussis* vaccines. Paper presentation at National Bioethics Conference, Deerfield, IL.

Pharmacy Experience

Advanced Community Pharmacy

April 2022

Indian Health Service, Albuquerque, NM

Preceptor: Khanh Vu, PharmD

- Presented updated asthma guidelines to clinical team to advocate guideline directed therapy
- Conducted anticoagulation, diabetes medication optimization, and smoking cessation appointments

Internal Medicine

March 2022

University of New Mexico Hospital, Albuquerque, NM

Preceptor: Ashley Fox, PharmD

- Researched and presented on Alcoholic Hepatitis
- Participated in clinical management of medications for inpatient internal medicine patients

Neonatal Intensive Care Unit

February 2022

University of New Mexico Children's Hospital, Albuquerque, NM

Preceptor: Eleni Shenk, PharmD

- Researched and presented on the nutrition supplementation in extremely premature neonates
- Participated in clinical management of medications for inpatient neonatal patients

Hematology Oncology

November 2021

University of New Mexico Hospital, Albuquerque, NM

Preceptor: Abigail Tresize, PharmD

- Researched and presented on the impact of small molecule targeted therapies on cancer outcomes over 20 years
- Participated in clinical management of medications for inpatient hematology oncology patients

Clinical Trials Research, Cooperative Studies Program

October 2021

Veterans Affairs, Office of Research and Development, Albuquerque, NM

Preceptor: Alexa Goldberg, PharmD, MS

- Improved knowledge regarding clinical trial design, ethics in clinical trials, and statistical methods
- Researched and presented on machine learning enabled healthcare technology using clinical trials

Infectious Disease

September 2021

University of New Mexico Hospital, Albuquerque, NM

Preceptor: Renee Mercier, PharmD

- Assessed the value of new medications in the role of HIV management
- Rounded with infectious disease consult team, recommended treatment regimens, and monitored patient progress
- Presented complex patient cases and microbiology results to state-wide ECHO conferences

Advanced Hospital Pharmacy

August 2021

Lovelace Westside Hospital, Albuquerque, NM

Preceptor: Stephanie Herrera, PharmD

- Rounded with the medical team and made recommendations for inpatient medication treatment
- Evaluated patient pharmacokinetic laboratory parameters and adverse effects for warfarin, vancomycin, and remdesivir to inform therapeutic drug monitoring
- Participated in supply chain management

Ambulatory Care: Anti-coagulation and Hepatitis C

July 2021

Christus St Vincent Hospital, Santa Fe, NM

Preceptor: Keith Romero, PharmD

- Conducted physical assessment and patient interviews
- Evaluated medication regimens for anti-coagulation patients on warfarin, apixaban, or rivaroxaban
- Assessed medication regimens for hepatitis C patients on Harvoni, Mavyret, or Epclusa
- Counseled patients on high risk anti-coagulation and antiviral medications

Pharmacy Medication Reconciliation

January-April 2021

Presbyterian Emergency Department, Albuquerque, NM

Preceptor: Tim Donlin, PharmD

- Interviewed emergency department patients regarding outpatient medications
- Updated patient electronic health records
- Alerted medical team to potential drug interactions

Pharmacy Medication Reconciliation

August-October 2020

University of New Mexico Hospital, Albuquerque, NM

Preceptor: Jeffery Gabaldon, PharmD

- Interviewed pediatric patients and their parents/guardians regarding outpatient medication regimens
- Updated patient electronic health records
- Alerted medical team to potential drug interactions

Hospital Pharmacy

July 2020

University of New Mexico Hospital, Albuquerque, NM

Preceptor: Scott Roach, PharmD

- Worked with pharmacists to monitor and adjust medication regimens, assisted in clinical duties including dispensing and verifying medications
- Assisted technicians in sterile compounding of hazardous and non-hazardous medications and in stocking medications throughout the hospital
- Introduced to hospital pharmacy supply chain management

Community Pharmacy

July 2019

Indian Health Services, Albuquerque, NM

Preceptor: Maria Apodaca, PharmD

- Conducted physical assessment and medication management for warfarin monitoring, diabetes, asthma, and smoking cessation
- Counseled patients on outpatient medications
- Exposed to supply chain management in the IHS system

Work Experience

Pharmacy Tutor

2019-2020

UNM College of Pharmacy, Albuquerque, NM

- Tutored pharmacy students weekly in their coursework
- Guided students to develop study skills and test preparation strategies

Walgreens Specialty Pharmacy Intern/Technician

2016-2019

Lovelace Medical Center, Albuquerque, NM

- Delivered and facilitate pharmacist consultation of discharge medications to the patient's bedside in the hospital
- Assisted in management of specialty medication cases

Graduate Assistant-Instructor

2015-2016

New Mexico State University, Las Cruces, NM

- Instructor for College Algebra and Calculus Labs
- Worked with other instructors to develop content for homework and exams
- Led study sessions for students to build skills and prepare for tests

Leadership Roles

Elected Educational and Academic Lead

2023-Present

Student Council of the International Society of Pharmacoepidemiology

- Develop first annual student research challenge inviting students to submit research proposals and compete for prizes and recognition
- Collaborate with society fellows to host pharmacoepidemiology seminars on a quarterly basis

Elected President of PharmD Class of 2022

2018-2022

UNM College of Pharmacy, Albuquerque, NM

- Elected by classmates to lead the executive team and class in initiatives to foster community, professional development, and service opportunities
- Led an executive team to set and collect dues from members, plan events, and fundraise for class gifts, graduation regalia, and a celebration upon graduation

- Represented the class of 2022 to the student pharmacists council before the dean of the college and other leadership

Elected President Student Pharmacist Council 2020-2021

UNM College of Pharmacy, Albuquerque, NM

- Led planning efforts for the Community Outreach Day 2020; worked with other leaders to coordinate COVID-safe community service activities in NM for over 200 students, such as flu shot clinics, COVID education, and food donation
- Created a website to move all student outreach training online
- Led an executive team to organize and streamline all activities put on by the 15 separate student organizations at the college of pharmacy
- Disseminated information and regulations related to fundraising and event planning
- Maintained a calendar of events for the college

Elected Student President, Honors Program 2014-2015

Trinity College, Deerfield, IL

- Work with faculty to organize quarterly events
- Plan a ceremony and reception to celebrate honors program graduates

Community Service

Pharmacy Community Service 2018-2021

Albuquerque, NM

- Administer COVID-19 vaccines
- Immunize patients at flu shot clinics
- Assess geriatric patients for fall risk and cognitive function, recommend appropriate follow-up measures including assistive devices and further screening

President Community Partnerships Cabinet 2012-2015

Trinity College, Deerfield, IL

- Planned annual events with a cabinet of ten peers to offer opportunities for on-campus students to have a meaningful impact on the Chicago area through service partnerships such as Habitat for Humanity, nursing home visits, and homeless shelter visits
- Recruited, interviewed, and trained new leaders for sub-groups
- Managed permission forms, school van certifications and rentals, and budget

President Habitat for Humanity 2013-2014

Trinity College, Deerfield, IL

- Planned six annual events with a cabinet of peers raise to funds and awareness for Lake County Habitat for Humanity
- Delegated responsibilities for fundraising, advertising, and bookkeeping

Awards and Accomplishments

First Place Team-Clinical Research Challenge 2019

UNM College of Pharmacy, Albuquerque, NM

- Competed with a team of two other students to win the American College of Clinical Pharmacy's local competition
- Advanced to final round of national competitions and placed among top seven teams in the country

Dean's Honor Role 2018-2022

UNM College of Pharmacy, Albuquerque, NM

Academic Scholarship Award 2018-2022

UNM College of Pharmacy, Albuquerque, NM

- Awarded to pharmacy students for outstanding academic achievements
- \$3,000 award, received annually

William C. Fiedler Scholarship

2019

UNM College of Pharmacy, Albuquerque, NM

- Awarded to pharmacy students for academic achievement and community service
- \$1,000 award

Ghattas Family Scholarship

2018

UNM College of Pharmacy, Albuquerque, NM

- Given to pharmacy students for academic achievement and community service
- \$1,000 award

Licenses

New Mexico Pharmacy License RP00009814

2022-Present

Pennsylvania Pharmacy License RP457276

2022-Present

**Professional
Memberships**

International Society of Pharmracoepidemiology

2022-Present

Society for Epidemiologic Research

2022-Present

American Pharmacists Society

2021-Present